

Name: _____

Date: _____

Chapter 9 Quiz1. Determine the oxidation number of each element in $\text{Cu}_3(\text{PO}_4)_2$ (3 marks)

	Oxidation Number
Cu	+2 ✓
P	+5 ✓
O	-2 ✓

$$1(\text{P}) + 4(-2) = -3$$

$$\text{P} = 5$$

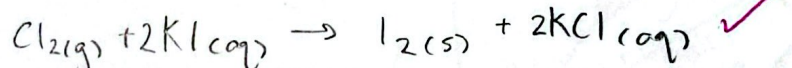
$$3(\text{Cu}) + 2(-3) = 0$$

$$\frac{3\text{Cu}}{3} = \frac{6}{3}$$

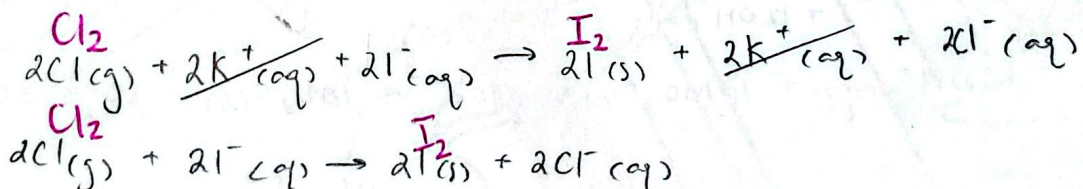
$$\text{Cu} = 2$$

2. Chlorine gas reacts with a potassium iodide solution to produce solid iodine in a solution of potassium chloride.

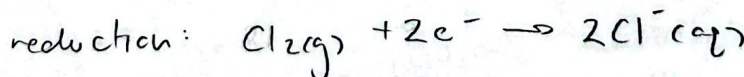
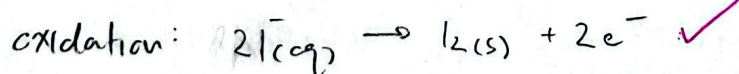
a) Write the balanced chemical equation for this reaction. (1 mark)



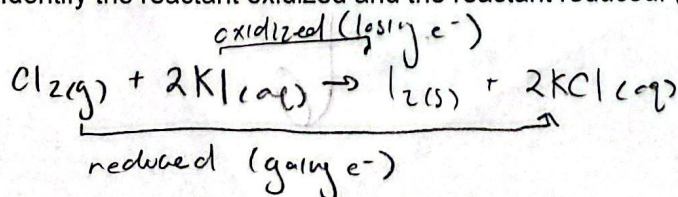
b) Write the net ionic equation for the reaction. (2 marks)



c) Write the oxidation and reduction half-reaction equations for the reaction. (2 marks)

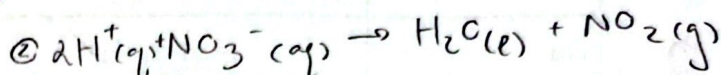
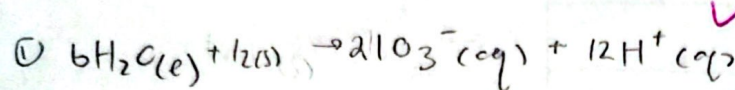
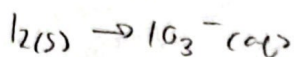
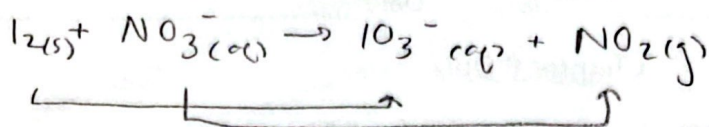


d) Identify the reactant oxidized and the reactant reduced. (1 mark)



∴ Cl is being reduced ✓
I is being oxidized ✓

3. Balance the following redox reaction in a **basic** solution. Use either the half reaction method or the oxidation number method in your solution: $I_{2(s)} + NO_3^-(aq) \rightarrow IO_3^-(aq) + NO_{2(g)}$ (6 marks)



RC: 0

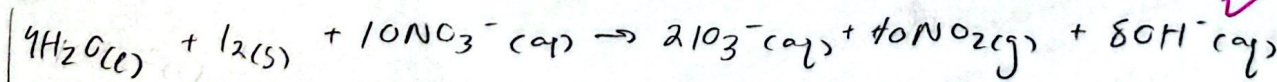
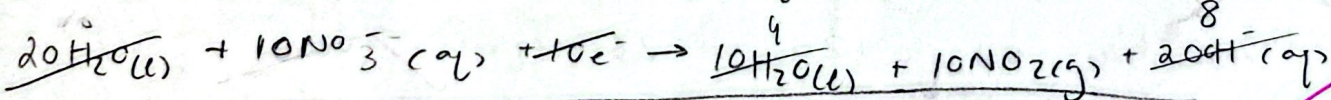
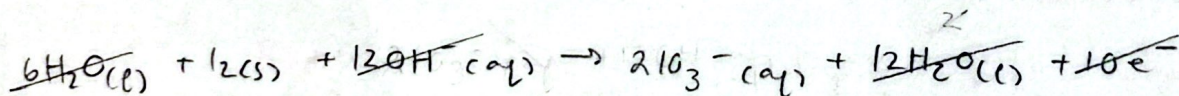
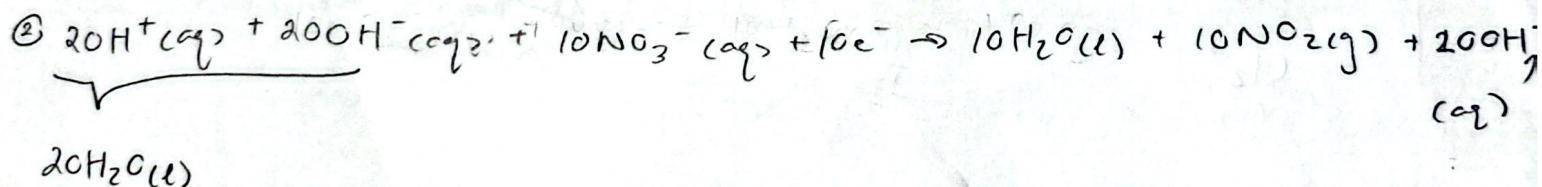
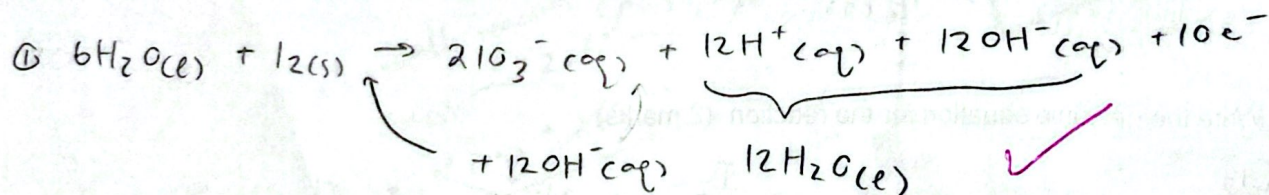
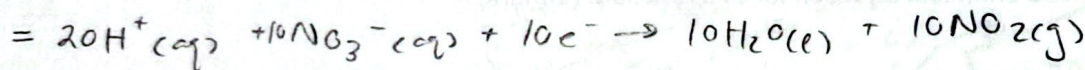
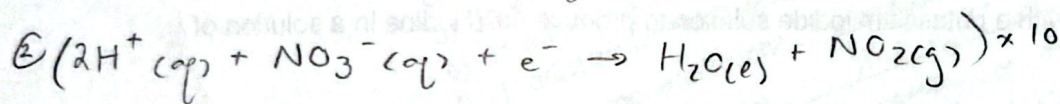
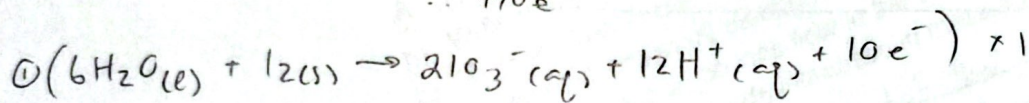
$$2(-1) + 12(+1) = +10$$

$$\therefore +10e^-$$

$$T/C: 2(+1) + 1(-1) = +1$$

0

$$\therefore +1e^-$$



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